

2016.0 RANGE ROVER (LG), 501-27

FRONT END SHEET METAL REPAIRS

FRONT SIDE MEMBER SECTION (G1454588)

REMOVAL AND INSTALLATION

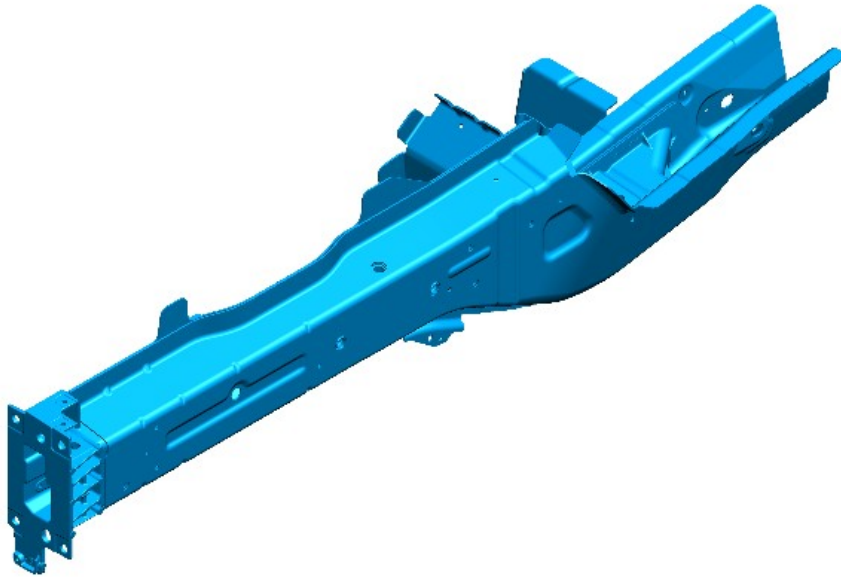
REMOVAL

NOTE:

The front side member section is installed in conjunction with:

- Front bumper cover
 - Front bumper
 - Front fender
 - Fender splash shield
 - Hood latch panel
 - Air cleaners
 - Radiators
 - Charge air cooler
 - Front suspension
 - Front subframe
 - Anti-lock brake system (ABS) module
 - Engine
-

1.



E143199

The front side member section is serviced as indicated.

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2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements.

For additional information, refer to: [Health and Safety Precautions](#) (100-00 General Information, Description and Operation).

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3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

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4. To make sure the vehicle is correctly aligned. It must be placed on a Land Rover approved alignment jig.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.

- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

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5. For additional information, refer to: [Front Bumper Cover](#) (501-19 Bumpers, Removal and Installation).
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6. For additional information, refer to: [Front Bumper](#) (501-19 Bumpers, Removal and Installation).
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7. For additional information, refer to: [Fender](#) (501-02 Front End Body Panels, Removal and Installation).
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8. For additional information, refer to: [Fender Splash Shield](#) (501-02 Front End Body Panels, Removal and Installation).
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9. For additional information, refer to: [Hood Latch Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).
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10.

NOTE:

Left hand only.

For additional information, refer to: [Left Air Cleaner](#) (303-12A Intake Air Distribution and Filtering - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation) / [Left Air Cleaner](#) (303-12E Intake Air Distribution and Filtering - TDV8 4.4L Diesel, Removal and Installation) / [Left Air Cleaner](#) (303-12C Intake Air Distribution and Filtering - V8 N/A 5.0L Petrol, Removal and Installation) / [Left Air Cleaner](#) (303-12D Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Removal and Installation).

11.

NOTE:

Right hand only.

For additional information, refer to: [Right Air Cleaner](#) (303-12A Intake Air Distribution and Filtering - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation) / [Right Air Cleaner](#) (303-12E Intake Air Distribution and Filtering - TDV8 4.4L Diesel, Removal and Installation) / [Right Air Cleaner](#) (303-12C Intake Air Distribution and Filtering - V8 N/A 5.0L Petrol, Removal and Installation) / [Right Air Cleaner](#) (303-12D Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Removal and Installation).

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12. For additional information, refer to: [Radiator](#) (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation) / [Radiator](#) (303-03F Engine Cooling - TDV8 4.4L Diesel, Removal and Installation) / [Radiator - V8 N/A 5.0L Petrol](#) (303-03C Engine Cooling - V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol, Removal and Installation) / [Radiator - V8 S/C 5.0L Petrol](#) (303-03C Engine Cooling - V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol, Removal and Installation).

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13. For additional information, refer to: [Condenser - TDV6 3.0L Diesel](#) (412-01A Climate Control, Removal and Installation) / [Condenser - TDV8 4.4L Diesel](#) (412-01A Climate Control, Removal and Installation) / [Condenser - V8 N/A 5.0L Petrol](#) (412-01A Climate Control, Removal and Installation).

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14. For additional information, refer to: [Shock Absorber and Spring Assembly](#) (204-01 Front Suspension, Removal and Installation).

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15. For additional information, refer to: [Front Subframe - V6 S/C 3.0L Petrol](#) (502-00 Uni-Body, Subframe and Mounting System, Removal and Installation) / [Front Subframe - TDV6 3.0L Diesel](#) (502-00 Uni-

Body, Subframe and Mounting System, Removal and Installation) / [Front Subframe - TDV8 4.4L Diesel](#) (502-00 Uni-Body, Subframe and Mounting System, Removal and Installation).

16.

NOTE:

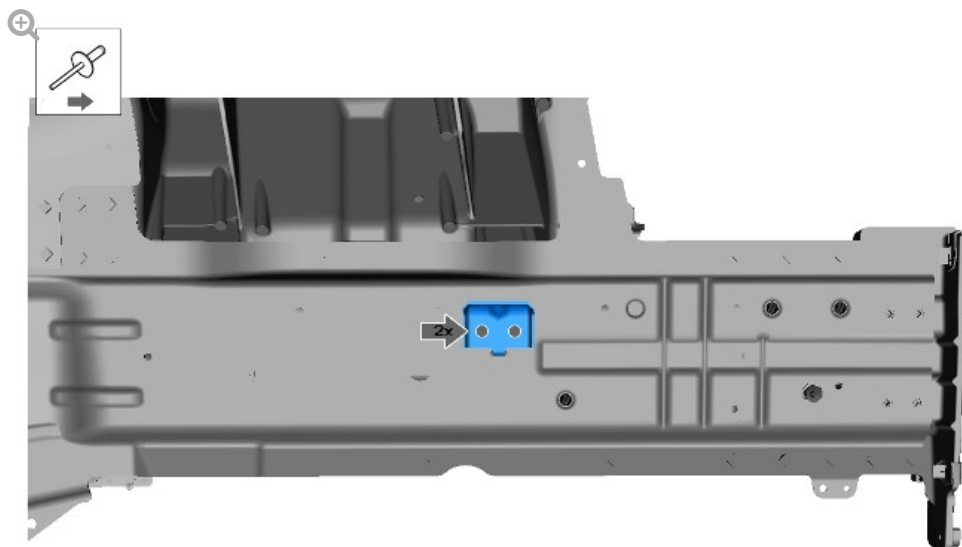
Right hand only.

For additional information, refer to: Anti-Lock Brake System (ABS) Module - RHD (206-09 Anti-Lock Control - Traction Control, Removal and Installation).

17. For additional information, refer to: [Engine](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal) / [Engine](#) (303-01E Engine - TDV8 4.4L Diesel, Removal) / [Engine](#) (303-01C Engine - V8 N/A 5.0L Petrol, Removal) / [Engine](#) (303-01D Engine - V8 S/C 5.0L Petrol, Removal).

18. Release the front side member wiring harness and position to one side.

19.

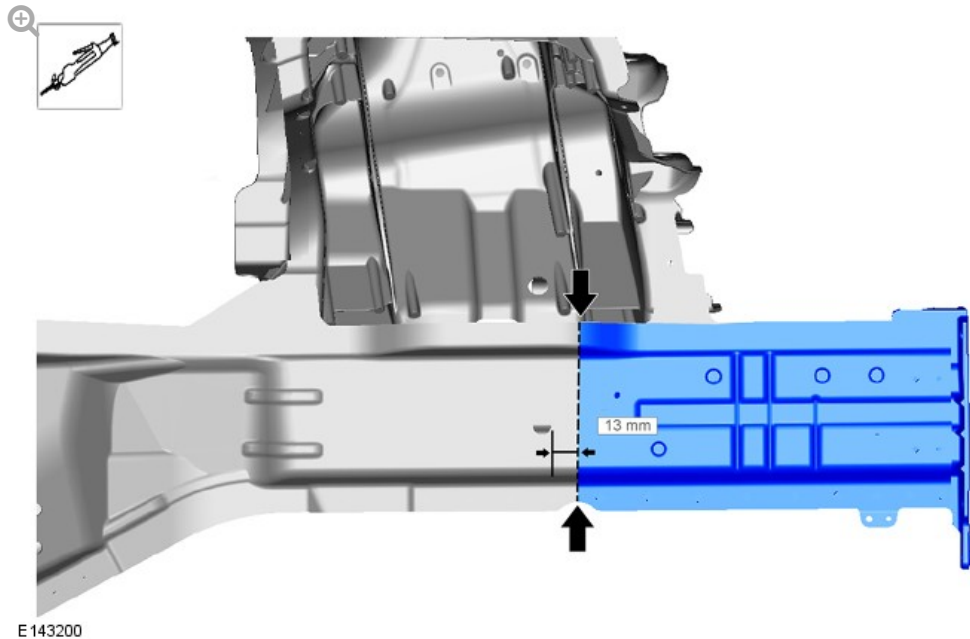


Remove the brake pipe retaining bracket as indicated.

20.

CAUTION:

The front side member and front side member closing panel joints are staggered.

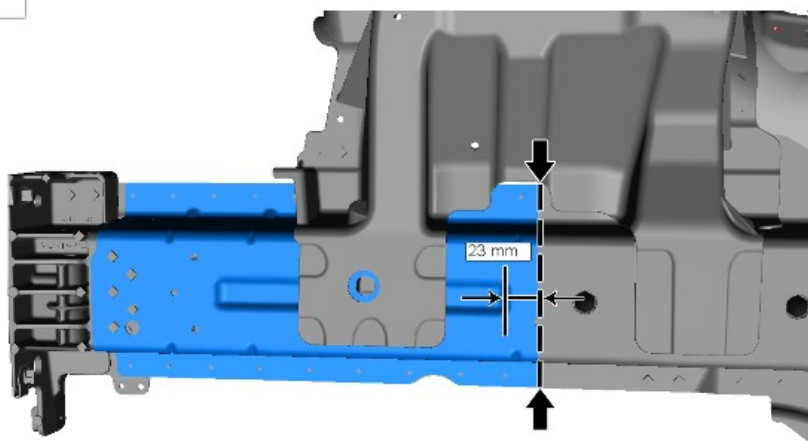


Mark out the position where the front side member closing panel section MIG (metal inert gas) butt joint is to be made. Cut through the front side member closing panel at this point, as indicated.

21.

CAUTION:

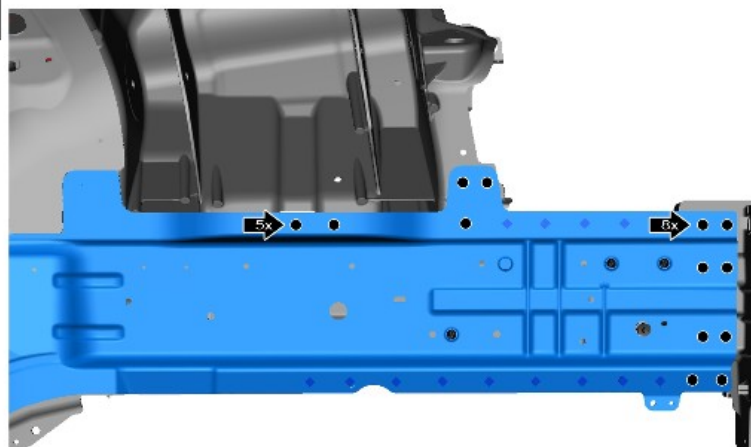
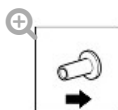
Care should be taken not to cut through into the front side member closing panel, as this section of the front side member closing panel remains in place.



E143201

Mark out the position where the front side member section MIG butt joint is to be made. Cut through the front side member at this point, as indicated.

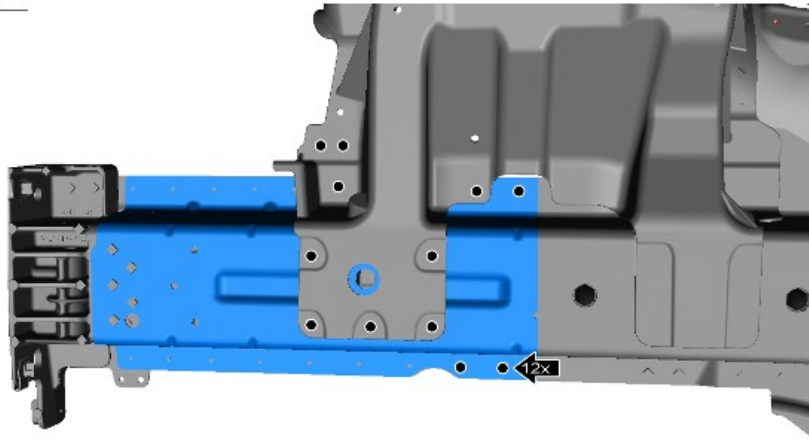
22.



E143266

Remove the SPR's from the front side member as indicated.

23.



E144548

Remove the SPR's from the front side member as indicated.

24. Separate the joints and remove the front side member section.

INSTALLATION

CAUTION:

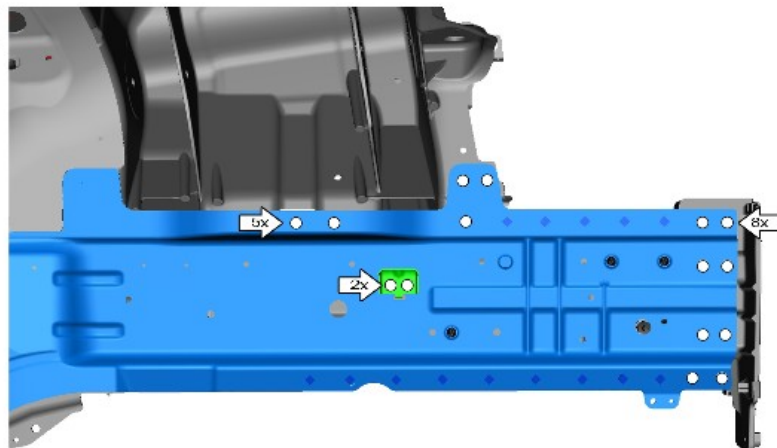
Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Land Rover approved SPR tool.

1. Remove the rivet remnants.

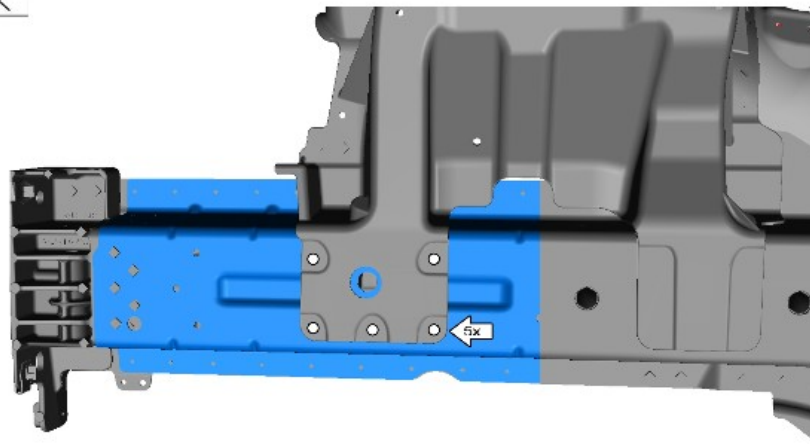
2. Dress the flanges where necessary.
3. Mark out the position on the front side member service panel where the MIG butt joint is to be made and cut the panel at this point.
4. Mark out the position on the front side member closing panel where the MIG butt joint is to be made and cut the panel at this point.
5. Offer up the new front side member service panel and clamp into position. Check alignment if correct proceed to the next step, if not rectify and recheck before proceeding.
6. Offer up the new front side member closing panel and clamp into position. Check alignment if correct proceed to next step, if not rectify and recheck before proceeding.
- 7.



E 143206

Drill holes at the points where BSF's are to be installed as indicated.

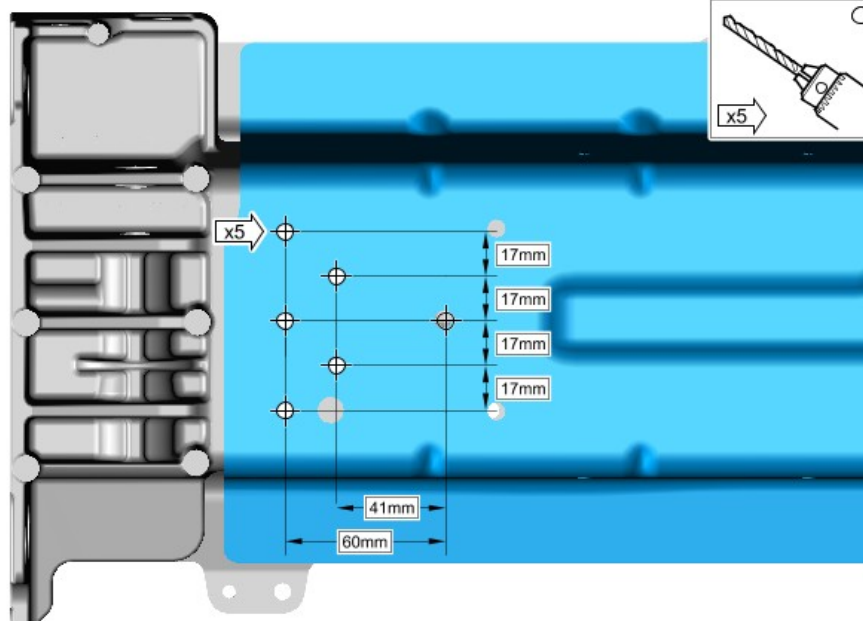
8.



E143265

Drill holes at the points where BSF's are to be installed as indicated.

9.



E178384

Mark, measure and drill holes where the BSF's are to be installed as indicated.

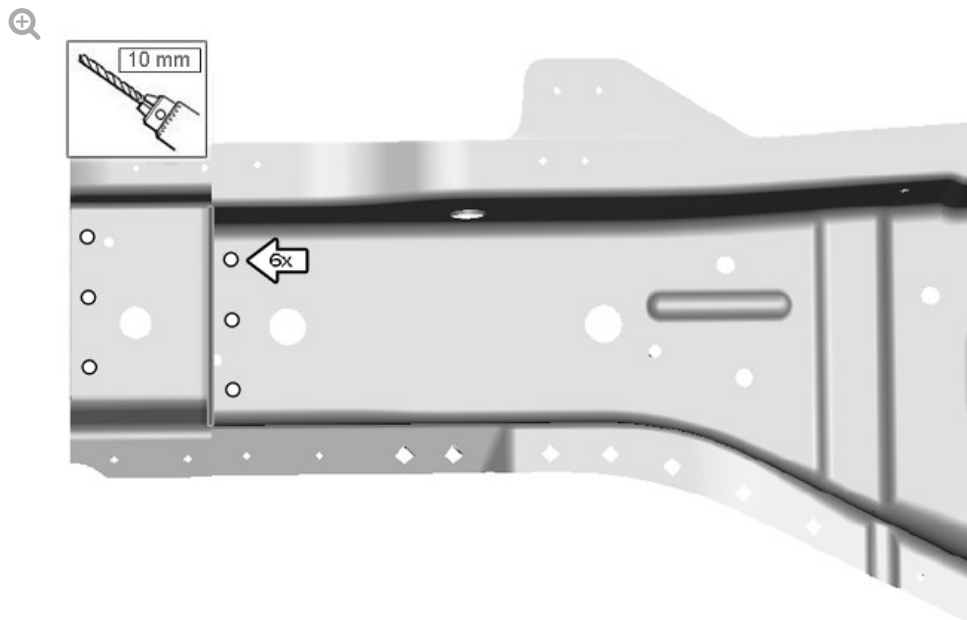
10. Deburr the drilled holes.

11. Using a fine bristle disc clean and prepare the panel surfaces including the inner panel surfaces where backing plates are to be

welded.

12. Measure and cut two 50mm wide backing plates, to be placed 25mm each side of the MIG butt joint from the discarded part of the new front side member panel.
13. Cut four run on/run off tabs from the discarded part of the new front side member remnant.
14. Using a fine bristle disc clean and prepare the backing plates the new front side member closing panel and new front side member panel.

15.



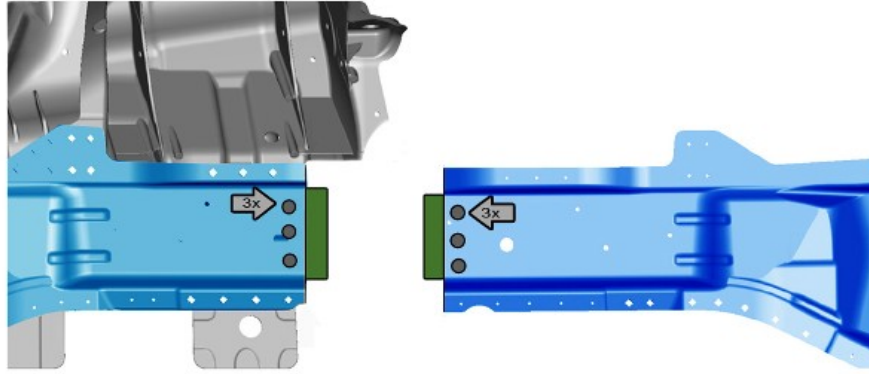
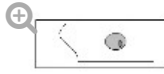
E143203

Drill holes in the vehicle assembly for the backing plates that are MIG plug welded as indicated.

16. Deburr the drilled holes.
17. Using a fine bristle disc clean and prepare the panel surfaces including the inner panel surfaces where backing plates are to be welded.

18. Offer up the backing plates, align and clamp into position. Check alignment if correct proceed to the next step, if not rectify and recheck before proceeding.

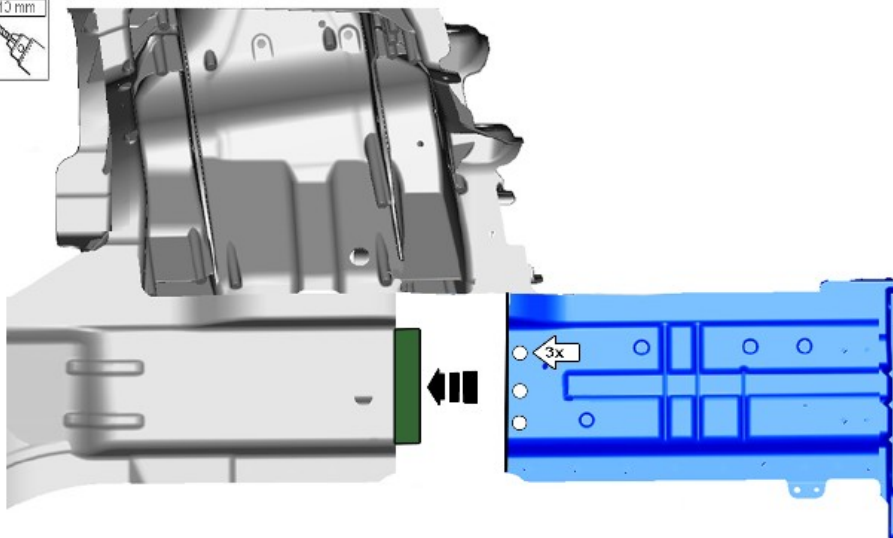
19.



E143204

MIG plug weld the backing plates to the vehicle assembly as indicated.

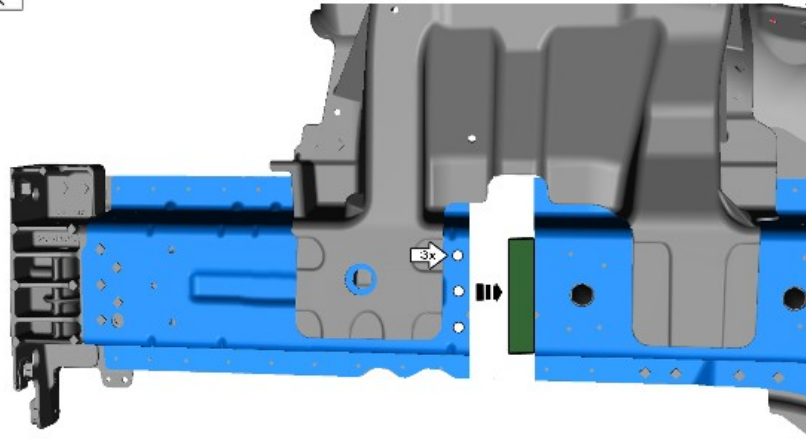
20.



E143208

Drill holes for the MIG plug welds for the installation of the backing plate to the front side member outer section as indicated.

21.



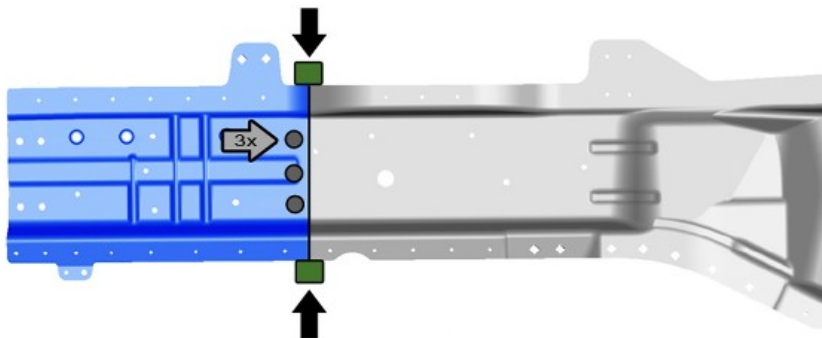
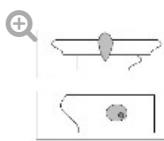
E143684

Drill holes for MIG plug welds for the installation of the backing plate to the front side member inner section as indicated.

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22. Offer up the new front side member closing panel to the vehicle, align and clamp into position. Check alignment if correct proceed to next step, if not rectify and recheck before proceeding.
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23. Tack weld the run-on/run-off tabs to the front side member section.
-

24.



E143686

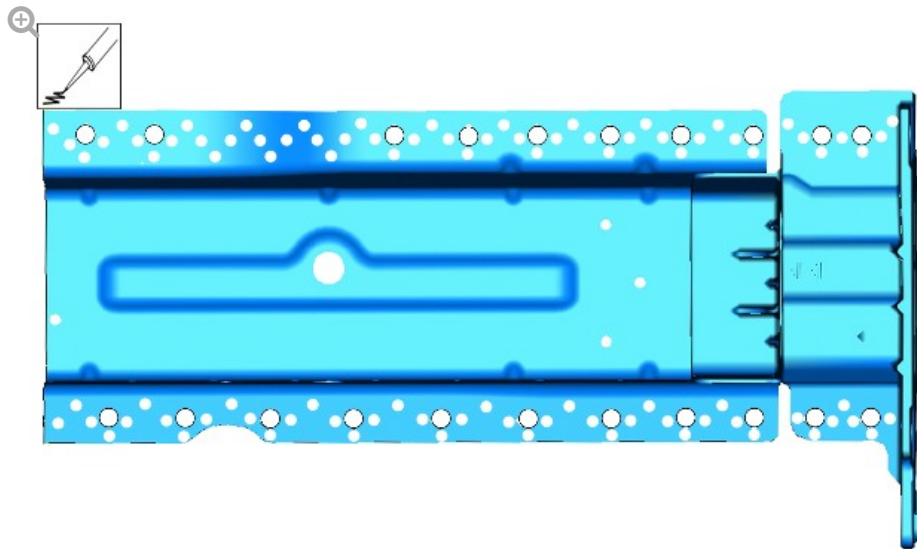
MIG plug weld the backing plates and MIG butt weld the front side member outer panel as indicated.

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25. Apply the coupling agent where the Land Rover recommended bonding material is to be applied and allow to dry.
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26.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.

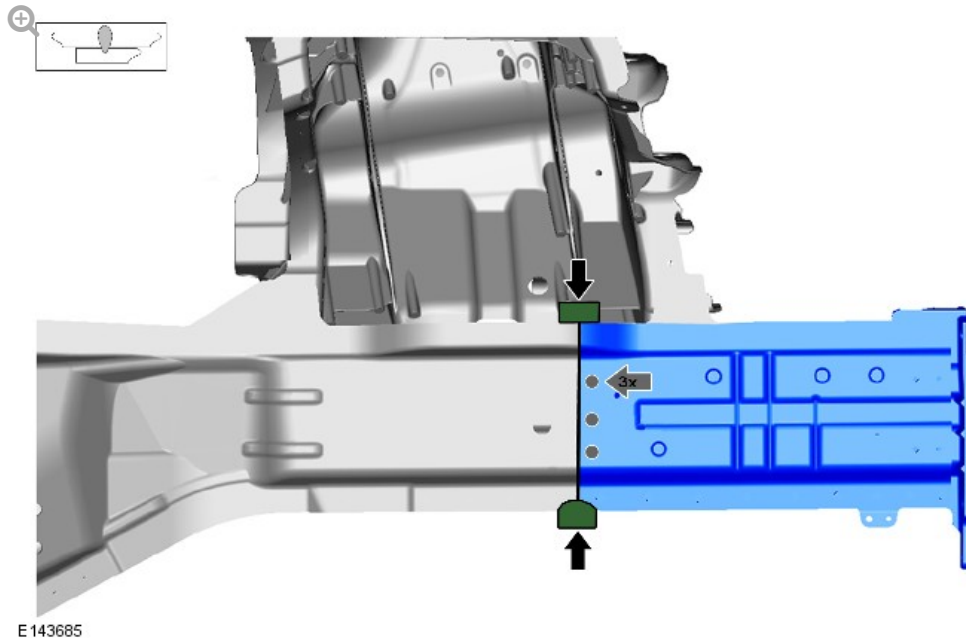


E143207

Apply a 5mm zig zag bead of Land Rover recommended bonding material to the front side member section.

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27. Offer up the new front side member closing panel to the vehicle, align and clamp into position. Check alignment if correct proceed to next step, if not rectify and recheck before proceeding.
-
28. Tack weld the run-on/run-off tabs to the front side member closing panel.
-

29.



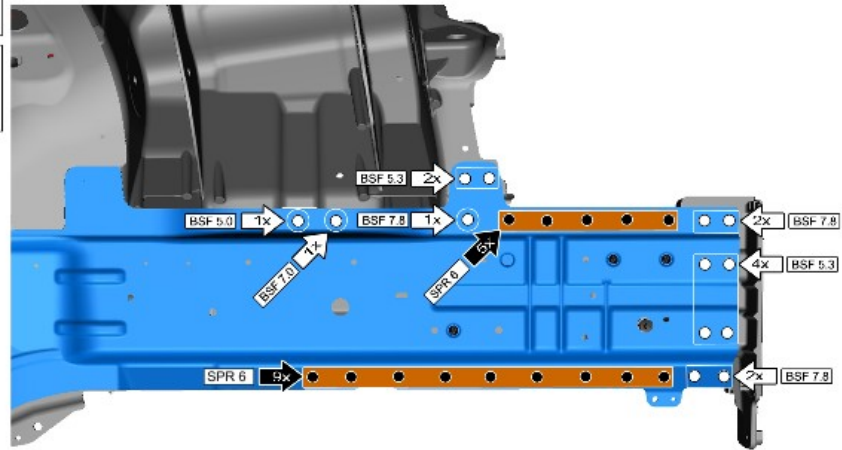
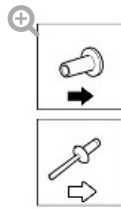
MIG plug weld the backing plates and MIG butt weld the front side member closing panel as indicated.

30. Cut off the run on/run off tabs.

31. Carry out a non destructive crack inspection on the MIG butt joints.
If correct proceed to next step if not, rectify and recheck before proceeding.

32. Dress the welded joints.

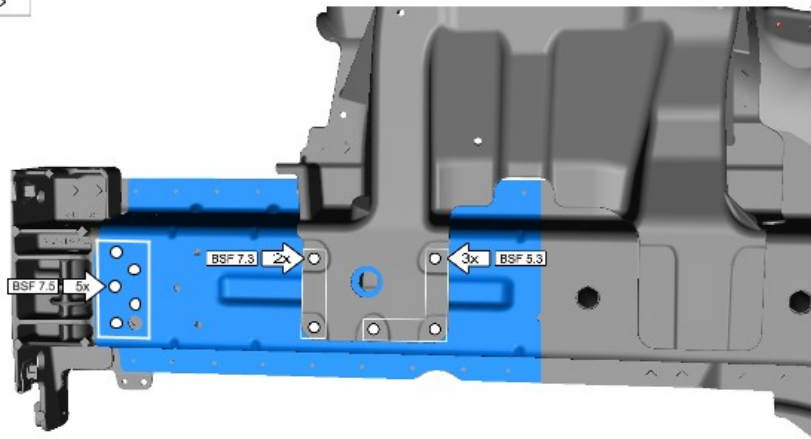
33.



E143205

Install the SPR's and the BSF's as indicated.

34.



E143687

Install the BSF's as indicated.

35. Make sure corrosion protection is applied to all areas affected by repair.

36. Make sure that any open or exposed panel joints are correctly sealed.

37. The installation of associated panels and components is the reversal of the removal procedure.